

Setup

The fundamental result we rely on is the Bolzano–Weierstrass theorem, which says every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

We also use the special case the mean value theorem for continuous functions on closed intervals.

Application

The optimisation argument splits into two halves. The first half appeals to *the Bolzano–Weierstrass theorem (p. 1)* to extract a convergent subsequence from the candidate set, then the second half uses *the mean value theorem (p. 1)* to locate the maximiser.

Concretely, the algorithm is well-defined whenever the input space is bounded — exactly the condition *the Bolzano–Weierstrass theorem (p. 1)* needs.

For the proof outline, see also the *Setup (p. 1)* section above.

A look back

Three cross-reference styles appear in this document:

- inline anchor on a `<span>` carrying the term itself (*the Bolzano–Weierstrass theorem (p. 1)*, captured at character level)
- block anchor on a heading (*Application (p. 1)*)
- the anchor's text and the page number both come from CSS, so removing or moving a section updates every reference automatically on the next render pass.